

Nasdaq Nordic – FIXT 1.1 Transport Layer

Nordic Equities - INET

Nordic Equity Derivatives - NDTS

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Table of Contents

1 Document Version History 4

2 Overview 5

2.1 Document References 5

2.2 Definitions & Abbreviations 5

3 Session Overview 6

3.1 Encryption 6

3.2 Supported Message Types 6

3.3 Session Rejects 6

3.4 Possible Duplicates 6

3.5 Reject Messages 6

3.6 Not Supported Tags 6

3.7 On Behalf Ofs 7

3.8 Connectivity 7

4 Overview 8

4.1 Fields 8

4.1.1 Date and Time fields 8

4.1.2 Note about Special Characters 8

5 Session Level Messages 9

5.1 Standard Messages 9

5.1.1 Standard Header 9

5.1.2 Standard Trailer 11

5.2 Reject 11

5.3 Heartbeat 13

5.4 Logon 13

5.5 Test Request 13

5.6 Resend Request 14

5.7 Sequence Reset 14

5.8 Logout 14

6 Appendix A – Max Lengths 15

6.1 Max Lengths - INET 15

6.2 Max Lengths - NDTS 15

Document Version History

Version	Date	Author	Summary of Changes
1.00	6 Dec 2016		Initial version
1.00.1	13 Mar 2017		Added Other to SessionRejectReason
1.00.2	1 Feb 2018		Added Trader ID to SenderSubID[50], TargetSubID [57] and OnBehalfOfSubID [116] Added reject reasons to SessionRejectReason [373] and corresponding text to Text [58]
1.00.3	25 May 2021		Updated section 5.5 Test Request.
1.00.4	25 Oct 2021		Merged INET and NDTS FIXT specifications. The following is updated: • Updated text in section 5 Session Level Messages to reflect INET and NDTS (Nasdaq Derivatives Trading System) specific functionality for tag 49, 50 and 57. • Section 4.1: INET: Added value “S” to SenderSubID [50] and TargetSubID [57] for FIX Order messages.

Overview

This document contains the FIXT 1.1 protocol used by Nasdaq Nordic and Nasdaq Baltic[1], the functionality of FIX, and the user requirements for accessing Nasdaq Nordic FIX protocols. This document outlines the messages transmitted via FIX for entering orders and trade reports into Nasdaq Nordic. This document does not consider messages transmitted via other entry points into Nasdaq Nordic.

This document describes order entry message requirements between FIX users and the back-end FIX engine.

Users should be familiar with the FIXT 1.1 specifications, as this document supplements those specifications.

2.1 Document References

Specification documents can be located at:

- https://www.nasdaq.com/solutions/european-trading-services
- https://memberportal.nasdaq.com/client/resources

It is assumed that the user of this manual is familiar with FIX 5.0 SP2 standard, which may be found at www.fixprotocol.org

2.2 Definitions & Abbreviations

"NDTS"	Nasdaq Derivatives Trading System
"INET"	Nasdaq Nordic Equities System

[1] Nasdaq Copenhagen, Nasdaq Helsinki, Nasdaq Iceland, Nasdaq Riga, Nasdaq Stockholm, Nasdaq Tallinn and Nasdaq Vilnius are respectively brand names for Nasdaq Copenhagen A/S, Nasdaq Helsinki Ltd, Nasdaq Iceland hf., Nasdaq Riga AS, Nasdaq Stockholm AB, Nasdaq Tallinn AS and AB Nasdaq Vilnius. Nasdaq Nordic represents the common offering by Nasdaq Copenhagen, Nasdaq Helsinki, Nasdaq Iceland and Nasdaq Stockholm. Nasdaq Baltic represents the common offering by Nasdaq Tallinn, Nasdaq Riga and Nasdaq Vilnius.

Session Overview

3.1 Encryption

Nasdaq Nordic does NOT support encryption through FIX.

3.2 Supported Message Types

Messages included in this specification are supported.

3.3 Session Rejects

Sometimes, messages that are sent inbound to the FIX engine are rejected directly resulting in session level rejects. In these cases, you will receive a reject message (35=3) that includes the reject reason (tag 373) and detailed, specific causes of the reject (tag 58). For more details please see section 5.2.

3.4 Possible Duplicates

You should set tag 43, PossDupFlag, if you are sending a message with a sequence number that was already used. This tag is required for retransmissions. The FIX engine will process as follows:

If tag 43 =	& Seq # is a dup	Then, the FIX engine will...
Y	Y	Ignore the message.
Y	N	Process the message normally.
N	Y	Drop the connection.
N	N	Process the message normally.

3.5 Reject Messages

Messages can be rejected when tag content validation fails, resulting in business level rejects. Messages can also be rejected by the application. Both business and application rejects include trade report, order, cancel, and cancel/replace rejects.

3.6 Not Supported Tags

If you send an unsupported tag, the tag and corresponding data will be ignored. Your message will not be rejected.

3.7 On Behalf Ofs

The FIX engine allows service bureaus to submit transactions for several customers over a single connector. These customers enter the application as their own entity, that is, with their own MPID. Service bureaus are not required to have a distinct connector for each customer. A service bureau may, however, have multiple connectors and use them however they wish.

Service bureaus will be required to submit a message containing:

- Tag 49, SenderCompId, the service bureau's MPID in INET, the service bureau's FIX Session ID in NDTS;
- Tag 115, OnBehalfOfId, the individual customer's MPID; and
- Tag 116, OnBehalfOfSubID, the User ID (Trader ID) for the customer

The processing of these tags for service bureaus differs from standard processing. The SenderCompID submitted in tag 49 identifies the service bureau. It could be a valid MPID or a representative code used for FIX processing only. This flexibility is allowed because the value is not passed to a destination system. Tag 115 identifies the customer's MPID.

Example:

Service bureau ABCD with TraderID A1234 is sending an order for customer EFGH with TraderID E5678.

The header inbound to the FIX engine contains the following:

- Tag 49 SenderCompID = ABCD
- Tag 50 SenderSubID = A1234. Optional and ignored by the FIX engine.
- Tag 115 OnBehalfOfCompID= EFGH
- Tag 116 OnBehalfOfSubID = E5678

When a response is returned from the FIX engine back to the customer, the header will contain the following:

- Tag 56 TargetCompID = ABCD
- Tag 128 DeliverToCompID = EFGH
- Tag 129 DeliverToSubID = E5678
- Note that Tag 57 = TargetSubID is not provided

3.8 Connectivity

- The initial logon message is validated for both the SenderCompID and TargetCompID as provided by Nasdaq.
- If logon fails, you will not receive a message indicating this failure.

Overview

4.1 Fields

4.1.1 Date and Time fields

All date and time fields for Nasdaq Nordic are to be sent in UTC (Universal Timestamp Coordinated) physical format expressed as GMT.

UTC Format is:

- YYYYMMDD-HH:MM:SS (whole seconds) format, colons, dash, and period required;

or

- YYYYMMDD-HH:MM:SS.sssssssss (subseconds down to nanoseconds) format, colons, dash, and period required. Although you may send subseconds, this is optional.

4.1.2 Note about Special Characters

Special characters (non-alphanumeric characters) should not be used in any fields (with the exception of the decimal point, in price fields). If special characters are used, Nasdaq reserves the right to replace these characters with a space. As a result, your outbound messages might contain values other than what you sent in. To avoid this occurrence, customers are advised to refrain from using any non-alphanumeric characters.

Session Level Messages

5.1 Standard Messages

5.1.1 Standard Header

Tag	Field	Reqd	Comments
8	BeginString	Y	FIXT.1.1 (Always unencrypted, must be first field in message)
9	BodyLength	Y	(Always unencrypted, must be second field in message)
35	MsgType	Y	(Always unencrypted, must be third field in message)
34	MsgSeqNum	Y	Integer message sequence number. Sequence numbers are initialized at the start of each FIX session starting at 1 (one) and increment throughout the session.
43	PossDupFlag	N	Always required for retransmitted messages, whether prompted by the sending system or as the result of a resend request. Indicates possible retransmission of message with this sequence number Valid values: N = Original transmission Y = Possible duplicate
49	SenderCompID	Y	Inbound to Nasdaq: SenderCompID as assigned by Nasdaq Nordic (MPID in INET, FIX Session ID in NDTS). Outbound from Nasdaq: From 56 on the original inbound message.
50	SenderSubID	N	Inbound to Nasdaq: Trader ID (User ID). This tag is not required on session level messages but is required for application level messages. Outbound from Nasdaq: INET: Indicator to denote which host system was involved: S = Order message T = Trade Reporting message NDTS: N/A

Tag	Field	Reqd	Comments
52	SendingTime	Y	UTC Outbound from Nasdaq: Please note that this time is the time the message left the FIX application. It has no relevancy to execution time, reject time, or cancel time in the destination systems.
56	TargetCompID	Y	Inbound to Nasdaq: As per the information provided by Nasdaq Nordic. Outbound from Nasdaq: From 49 on the original inbound message.
57	TargetSubID	N	Inbound to Nasdaq: INET: Required 1-character that identifies this entry as an order entry or trade reporting message. S = Order message T = Trade Reporting message NDTS: Optional indicator to denote client software version. Outbound from Nasdaq: Receiving party's Trader ID (User ID), if this message is a query/response message. From 50 on the original inbound message. Not specified for On-Behalf-Of message responses. This tag is not required on session level messages.
97	PossResend	N	Inbound to Nasdaq: Ignored Outbound from Nasdaq: N/A
115	OnBehalfOfCompID	N	Inbound to Nasdaq: Service Bureaus use this field to submit their customer's MPID. This value will be returned to you in tag 128 on the outbound message. Please note that while it is possible for you submit a chain of transactions with inconsistent values for this tag in different messages, you are strongly advised not to do so. Outbound from Nasdaq: N/A
116	OnBehalfOfSubID	N	Inbound to Nasdaq: When applicable, this tag holds the User ID (Trader ID) for the on behalf of firm. Outbound from Nasdaq: N/A

Tag	Field	Reqd	Comments
122	OrigSendingTime	N	Required for message resent as a result of a ResendRequest. If data is not available set to same value as SendingTime UTC
128	DeliverToCompID	N	Inbound to Nasdaq: N/A Outbound from Nasdaq: For drop and service bureau on behalf of messages, this field contains the service bureau's customer's MPID.
129	DeliverToSubID	N	Inbound to Nasdaq: N/A Outbound from Nasdaq: For drop and service bureau on behalf of messages, this field contains the value entered in tag 116 on the inbound message.
145	DeliverToLocationID	N	Inbound to Nasdaq: N/A Outbound from Nasdaq: For drop messages, this field (if provided) specify the account / port ID used when originally sending the message. Use together with OrigClOrdID [41] and DeliverToCompID [128] to chain an order modification to the original order.

5.1.2 Standard Trailer

Tag	Field	Reqd	Comments
10	CheckSum	Y	

5.2 Reject

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 3
45	RefSeqNum	Y	MsgSeqNum of rejected message
58	Text	N	Where possible, coded message that includes an indicator and text to explain reason for rejection: Invalid tag number; Required tag missing; Tag not defined for this message type; Undefined Tag; Tag specified without a value; Value is incorrect (out of range) for tag; Incorrect data format for value; Signature problem; Tag CompID problem; SendingTime accuracy problem; Invalid value for

Tag	Field	Reqd	Comments
			MsgType tag; Repeating group fields out of order; Incorrect number in group count for repeating group
371	RefTagID	N	The tag number of the FIX field being referenced.
372	RefMsgType	N	The MsgType of the FIX message being referenced.
373	SessionRejectReason	Y	Code to identify reason for a session-level Reject message. Valid values: 0 = Invalid tag number 1 = Required tag missing 2 = Tag not defined for this message type 3 = Undefined tag 4 = Tag specified without a value 5 = Value is incorrect (out of range) for this tag 6 = Incorrect data format for value 7 = Decryption problem 8 = Signature problem 9 = CompID problem 10 = SendingTime accuracy problem 11 = Invalid MsgType 15 = Repeating group fields out of order 16 = Incorrect number in group count for repeating group 99 = Other
	StandardTrailer	Y	

5.3 Heartbeat

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 0
112	TestReqID	N	Required when the heartbeat is the result of a Test Request message. It is the TestReqID received from the Test Request message.
	StandardTrailer	Y	

5.4 Logon

After establishing your TCP connection with Nasdaq FIX, a Logon must be the first message that you send. Nasdaq will respond with its own Logon. You must wait for that Logon before sending additional messages.

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = A
98	EncryptMethod	Y	Not supported. Must be 0. Valid values: 0 = None
108	HeartBtInt	Y	Heartbeat Interval in seconds– should be greater than 10 and less than or equal to 60.
1137	DefaultApplVerID	Y	The default version of FIX messages used in this session. Valid values: 9 = FIX50SP2
	StandardTrailer	Y	

5.5 Test Request

The HeartBtInt that is specified in the Logon message will be used to make sure that your connection is alive and functioning. If Nasdaq detects inactivity for a period longer than the HeartBtInt + 1 second, we will send a Test Request to determine if you are still active. There will be a total of three different Test Requests. On the next timeout, Nasdaq will immediately send a logout message and connection will be closed.

It is expected that similar logic will be used on your side in order to detect host activity.

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 1
112	TestReqID	Y	Date/Time combination
	StandardTrailer	Y	

5.6 Resend Request

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 2
7	BeginSeqNo	Y	Beginning sequence number of Requested Message
16	EndSeqNo	Y	Ending sequence number of Requested Message
	StandardTrailer	Y	

5.7 Sequence Reset

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 4
36	NewSeqNo	Y	Next expected sequence number
123	GapFillFlag	N	Valid values: Y = Gap Fill message, MsgSeqNum field valid
	StandardTrailer	Y	

5.8 Logout

Logouts will occur as a normal end of day routine but may also occur as a result of inactivity.

Tag	Field	Reqd	Comments
	StandardHeader	Y	MsgType = 5
58	Text	N	
	StandardTrailer	Y	

Appendix A – Max Lengths

6.1 Max Lengths - INET

Tag	Field	Max Length
49	SenderCompID	6
50	SenderSubID	6
56	TargetCompID	6
57	TargetSubID	6
115	OnBehalfOfCompID	6
116	OnBehalfOfSubID	6
128	DeliverToCompID	6
129	DeliverToSubID	6

6.2 Max Lengths - NDTS

Tag	Field	Max Length
49	SenderCompID	12
50	SenderSubID	6
56	TargetCompID	12
57	TargetSubID	6
115	OnBehalfOfCompID	5
116	OnBehalfOfSubID	6
128	DeliverToCompID	12
129	DeliverToSubID	6
145	DeliverToLocationID	20